

One truth across five channels

You are the incoming **Manager of Omnichannel Analytics**. Five channels report performance in five different systems, formats and definitions. Your job: turn fragmented data into **one governed, 360° commercial view** that Business Units and Brand teams can trust, self-serve and act on.

APPROACH

Independent work + present

TOOLS

SQL / Python / Power BI or anything preferred

SESSION

45 min live

SCOPE

5 channels · 6 markets

00 The context

Our region sells through **Retail, eCommerce (dotCom), the App, Franchise and Wholesale**. Today each channel reads performance separately — different systems, different KPI definitions and manual pulls. There is no single view of **what was ordered, how it sold, and how product performs** across categories and countries (AR, BR, CL, CO, MX, PE). Your case is to build the engine that closes that gap and to tell the story the numbers reveal.

01 Your source data — five raw extracts

CSV · RETAIL

Retail orders

~700K rows

retail_orders.csv

NDJSON · DOTCOM

eCommerce orders

~200K orders

ecomm_orders.ndjson

JSON · APP

App orders

~200K rows

app_orders.json

PARQUET · WHOLESALE

Wholesale orders

~48K rows

wholesale_orders.parquet

XLSX · FRANCHISE

Franchise orders

~45K rows

franchise_orders.xlsx

ALSO INCLUDED · CURRENCY REFERENCE

currency_reference.csv gives an exchange rate reference, in the column `local_units_per_eur` — the number of local-currency units that equal **1 EUR**. **EUR value = local amount ÷ local_units_per_eur**

Each file carries the same business concepts under **different names, encodings and levels of detail**. Read each source carefully — they don't all describe the world the same way or at the same grain, so part of the work is deciding how to make every figure **consistent and comparable** before you measure. Once mapped, the shared concepts are:

consumer id **channel** **country** AR/BR/CL/CO/MX/PE **year-month** **category** running/outdoor/football/originals **product id**
qty ordered & sold **price & currency** **order id** **adClub** Y/N **order status** processing / delivered / cancelled / returned

02 What to deliver

- 1 Data Layer.** A clean, documented, reproducible model that unifies the five channels into one governed dataset with a single consistent schema. Prepare a one pager with a roadmap that summarizes how you would handle the project, including if you would include any stakeholder to work with, why and in which phase.
- 2 Governed KPIs.** Define at least 3 standard commercial metrics with explicit, versioned definitions. State precisely **which records count as a real sale — and why**.
- 3 Executive view.** Build a clear visual / report / analytical product on the final dataset that translate the metrics into a digestible, C-level view of your choice.
- 4 Insight + story.** Go beyond the numbers: find one quantified commercial opportunity or pattern and turn it into a recommendation, in a **2-3 slide** executive readout with a clear call to action.
- 5 Governance & scale.** One final slide on how you'd turn this into a reusable tool and who you'd involve to sustain and adopt it.

03 Rules & how you're assessed

- Submit your **code + a complete README** about the whole project, steps applied and tools used + **the deck**. Bring them to the session.
- Treat the files as **real production extracts**. The figures are not real — they have been adapted for this exercise and do not reflect actual company results.
- State any assumptions if needed.** Explain why you're using them.
- 45-minute live session:** ~25 min presentation of your technical approach, the business connection and recommendations; ~20 min panel Q&A (technical and non-technical).
- Some questions can be asked in English**, be prepared.

What earns top marks: the **engine mindset** — reusable mapping logic, documented KPI definitions, attention to data quality, a governed single source of truth, and a story leadership would act on. A brilliant chart with no governance layer scores lower than a governed model with a clear so-what.

GOOD LUCK 😊